

$$A = (0, 2, 3) \quad B = (-1, 2, 3) \quad C = (1, 3, 4)$$

$$(1) \quad u = [\vec{AB}] = (-1, 0, 0)$$

$$v = [\vec{AC}] = (1, 1, 1)$$

$$\cos \varphi = \frac{\langle u, v \rangle}{\|u\| \|v\|} = \frac{-1}{1 \cdot \sqrt{3}} = -\frac{1}{\sqrt{3}} \Rightarrow \varphi = \arccos\left(-\frac{1}{\sqrt{3}}\right)$$

$$(2) \quad w = \vec{i} + \vec{j} + 2\vec{k}$$

$$u \times v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = +\vec{j} - \vec{k} = (0, 1, -1)$$

$$w' = w - \frac{\langle w, u \times v \rangle}{\|u \times v\|^2} (u \times v)$$

$$= (1, 1, 2) - \frac{(1-2)}{2} (0, 1, -1) =$$

$$= (1, 1, 2) + \frac{1}{2} (0, 1, -1) =$$

$$= \left(1, \frac{3}{2}, \frac{3}{2}\right)$$

$$(3) \quad V = \begin{vmatrix} -1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = |-1| = 1$$

$$U_1 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid (x-y)^2 + z^2 = 0 \right\}$$

$$= \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x-y=0 \quad z=0 \right\} = \left\{ \begin{pmatrix} x \\ x \\ 0 \end{pmatrix} \right\}$$

Siano $u = \begin{pmatrix} x \\ x \\ 0 \end{pmatrix}$ e $v = \begin{pmatrix} y \\ y \\ 0 \end{pmatrix}$ appartenenti a U_1

Dato lo scalare c , per provare che U_1 è sottospazio di $\mathbb{R}^3$ si verifica se il seguente vettore $cu - v \in U_1$

$$c \begin{pmatrix} x \\ x \\ 0 \end{pmatrix} - \begin{pmatrix} y \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} cx-y \\ cx-y \\ 0 \end{pmatrix} \in U_1$$

perché la terza componente è nulla e le prime due sono uguali

dim $U_1 = 1$ perché $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ è base di U_1

$$U_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid (x-y)^2 + z^2 = 1 \right\} \text{ non è sottospazio}$$

contiene $(0,0,0)$

non è sottospazio

$$U_3 = \left\{ \begin{pmatrix} x \\ x \\ x \end{pmatrix} \in \mathbb{R}^3, x \in \mathbb{R} \right\}$$

Prendi $\begin{pmatrix} x \\ x \\ x \end{pmatrix}$ e $\begin{pmatrix} y \\ y \\ y \end{pmatrix} \in U_3$

$$c \begin{pmatrix} x \\ x \\ x \end{pmatrix} - \begin{pmatrix} y \\ y \\ y \end{pmatrix} = \begin{pmatrix} cx-y \\ cx-y \\ cx-y \end{pmatrix} \in U_3 \quad (\text{tutte le componenti sono uguali})$$

Quunque U_3 è sottospazio di dim 1

perché $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ è base di U_3

$$U_1 + U_3 = \left[\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right]$$

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{rank } B = 2$$

$$\begin{aligned} \dim(U_1 + U_3) &= 2 = \dim U_1 + \dim U_3 - \dim(U_1 \cap U_3) \\ &= 1 + 1 - \dim(U_1 \cap U_3) \end{aligned}$$

$$\Rightarrow 2 = 2 - \dim(U_1 \cap U_3)$$

$$\text{siccome } \dim(U_1 \cap U_3) = 0$$

$$U_1 + U_3 \text{ è somma diretta} \Rightarrow U_1 \oplus U_3$$

$$A = \begin{pmatrix} 0 & -1 & -2 \\ -1 & 6 & k^2 \\ -2 & 4 & 2 \end{pmatrix}$$

$$\text{Ker } A = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{array}{l} -y - 2z = 0 \\ -x + 6y + k^2 z = 0 \\ -2x + 4y + 2z = 0 \end{array} \right\}$$

$$\begin{vmatrix} 0 & -1 & -2 \\ -1 & 6 & k^2 \\ -2 & 4 & 2 \end{vmatrix} = \begin{vmatrix} -1 & k^2 \\ -2 & 2 \end{vmatrix} - 2 \begin{vmatrix} -1 & 6 \\ -2 & 4 \end{vmatrix}$$

$$\text{rango } A \geq 2 \quad \begin{vmatrix} -1 & 6 \\ -2 & 4 \end{vmatrix} = -4 + 12 \neq 0$$

$$\begin{aligned} &= -2 + 2k^2 - 2 \cdot 8 = \\ &= -18 + 2k^2 = -2(9 - k^2) \quad -18 + 2k^2 = 0 \Leftrightarrow k = \pm 3 \end{aligned}$$

Per $k = \pm 3$, $\text{rango } A = 2$ e dunque il sistema omogeneo associato ad A ammette soluzioni non banali
 cioè $(\text{Ker } A) = 1$

$$-x + 6y = -9z$$

$$-2x + 4y = -2z$$

$$x = \frac{\begin{vmatrix} -9z & 6 \\ -2z & 4 \end{vmatrix}}{8} = \frac{-36z + 12z}{8} = \frac{-24z}{8} = -3z$$

$$y = \frac{\begin{vmatrix} -1 & -9z \\ -2 & -2z \end{vmatrix}}{8} = \frac{2z - 18z}{8} = \frac{-16z}{8} = -2z$$

$$\text{Ker } A = \left[\begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} \right]$$

per $k \neq \pm 3$, $\text{range } A = 3$

$$\text{Ker } A = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \dim(\text{Ker } A) = 0$$

— 0 —
Sistema lineare

$$\begin{cases} -y - 2z = k+1 \\ -x + 6y + k^2 z = 4 \\ -2x + 4y + 2z = 1 \end{cases}$$

$$A = \begin{pmatrix} 0 & -1 & -2 \\ -1 & 6 & k^2 \\ -2 & 4 & 2 \end{pmatrix}$$

Se $k = \pm 3$, $\text{range } A = 2$

$$\text{range}(A|b) = \begin{pmatrix} 0 & -1 & -2 & k+1 \\ -1 & 6 & 9 & 4 \\ -2 & 4 & 2 & 1 \end{pmatrix} \quad \begin{matrix} \pm 3 \\ k+1 \end{matrix}$$

$$\begin{vmatrix} 0 & -1 & \pm 3+1 \\ -1 & 6 & 4 \\ -2 & 4 & 1 \end{vmatrix} = (-1+8) + \begin{matrix} 4(-4+12) \\ -2(-4+12) \end{matrix}$$
$$= 7 + \begin{matrix} 4 \cdot 8 = 32 \\ -2 \cdot 8 = -16 \end{matrix} \neq 0$$

$\text{range}(A|b) = 3 \neq \text{range } A$
sistema incompatibile

Se $k \neq \pm 3$, $\text{range } A = 3$, $\text{range}(A|b) = 3$

Il sistema è di Cramer.

Calcolo l'unica soluzione

$$\det A = -2(9 - k^2)$$

$$x = \frac{\begin{vmatrix} k+1 & -1 & -2 \\ 4 & 6 & k^2 \\ 1 & 4 & 2 \end{vmatrix}}{-2(9 - k^2)} = \frac{(k+1)(12 - 4k^2) + 8 - k^2 - 20}{-2(9 - k^2)}$$

$$= \frac{12k - 4k^3 + 12 - 4k^2 - 12 - k^2}{-2(9 - k^2)} = \frac{k(12 - 4k^2 - 5k)}{-2(9 - k^2)}$$

$$= \frac{+k(4k^2 + 5k - 12)}{+2(9 - k^2)}$$

$$y = \frac{\begin{vmatrix} 0 & k+1 & -2 \\ -1 & 4 & k^2 \\ -2 & 1 & 2 \end{vmatrix}}{-2(9 - k^2)} = \frac{-(k+1)(-2 + 2k^2) - 2(-1 + 8)}{-2(9 - k^2)}$$

$$= \frac{+(-2k + 2k^3 - 2 + 2k^2 + 14)}{+2(9 - k^2)} =$$

$$= \frac{(2k^3 + 2k^2 - 2k + 12)}{2(9 - k^2)}$$

$$z = \frac{\begin{vmatrix} 0 & -1 & k+1 \\ -1 & 6 & 4 \\ -2 & 4 & 1 \end{vmatrix}}{-2(9 - k^2)} = \frac{-1 - 4k - 4 - 2(-4 - 6k - 6)}{-2(9 - k^2)}$$

$$= \frac{-5 - 4k + 20 + 12k}{-2(9 - k^2)} = \frac{15 + 8k}{-2(9 - k^2)}$$

$$f: \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \rightarrow \begin{pmatrix} ax \\ y-t \\ 2x+az \end{pmatrix}$$

④. Si dimostra che f è lineare.
Prendi due elementi di $\mathbb{R}^4$,

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ t_1 \end{pmatrix} \text{ e } \begin{pmatrix} x_2 \\ y_2 \\ z_2 \\ t_2 \end{pmatrix} \text{ e}$$

scalari $\alpha, \beta \in \mathbb{R}$

$$f\left(\alpha \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ t_1 \end{pmatrix} + \beta \begin{pmatrix} x_2 \\ y_2 \\ z_2 \\ t_2 \end{pmatrix}\right) = f\begin{pmatrix} \alpha x_1 + \beta x_2 \\ \alpha y_1 + \beta y_2 \\ \alpha z_1 + \beta z_2 \\ \alpha t_1 + \beta t_2 \end{pmatrix} =$$

$$= \begin{pmatrix} a(\alpha x_1 + \beta x_2) \\ \alpha y_1 + \beta y_2 - \alpha t_1 - \beta t_2 \\ 2(\alpha x_1 + \beta x_2) + a(\alpha z_1 + \beta z_2) \end{pmatrix} =$$

$$= \begin{pmatrix} \alpha ax_1 + \beta ax_2 \\ \alpha (y_1 - t_1) + \beta (y_2 - t_2) \\ \alpha (2x_1 + az_1) + \beta (2x_2 + az_2) \end{pmatrix} =$$

$$= \alpha \begin{pmatrix} ax_1 \\ y_1 - t_1 \\ 2x_1 + az_1 \end{pmatrix} + \beta \begin{pmatrix} ax_2 \\ y_2 - t_2 \\ 2x_2 + az_2 \end{pmatrix} = \alpha f\begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ t_1 \end{pmatrix} + \beta f\begin{pmatrix} x_2 \\ y_2 \\ z_2 \\ t_2 \end{pmatrix}$$

$$(2) \quad \text{Ker } f = \text{Ker } A = \left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} : \begin{array}{l} ax = 0 \\ y - t = 0 \\ 2x + az = 0 \end{array} \right\}$$

$$\hookrightarrow A = \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 2 & 0 & a & 0 \end{pmatrix} \text{ matrice associata a } f$$

$$\text{rang } A \geq 2$$

$$\begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} = -2 \neq 0$$

$$\begin{vmatrix} a & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & a \end{vmatrix} = a^2$$

$$\begin{vmatrix} a & 0 & 0 \\ 0 & 1 & -1 \\ 2 & 0 & 0 \end{vmatrix} = 2 \cdot 0 = 0$$

Dunque, affinché $\text{rang}(A) = 3$ deve essere $a \neq 0$

Per $a = 0$, $\text{rang } A = 2$.

Imm ha dimensione 3 per $a \neq 0$ e 2 per $a = 0$

$$\begin{aligned} \text{Siccome } \dim(\text{Ker } f) &= 4 - \dim(\text{Im } f) \\ &= 4 - \text{rang}(A) \end{aligned}$$

$$\text{si ha che } 2 = \dim(\text{Ker } f) = 4 - 2$$

ovvero $\text{rang}(A) = 2$ e dunque $a = 0$.

$$(3) \quad \text{Per } a = 0 \quad \text{Ker } f = \left[\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right] \quad \dim(\text{Ker } f) = 2$$

funzione non
iniettiva e
non suriettiva

$$\text{per } a \neq 0 \quad \text{Ker } f = \left[\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right] \quad \dim \text{Ker } f = 1$$

funzione non
iniettiva ma
suriettiva

③ per $a = 0$

$$\text{Im} f = \left[\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right]$$

④ per verificare se $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \in \text{Im} f$ per $a = 0$
occorre verificare se

$B = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$ ha range 2. In caso contrario $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \notin \text{Im} f$

$$\begin{vmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{vmatrix} = 0$$

Dunque $\text{range}(B) = 2$
e $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \in \text{Im} f$.

⑤ determino $f^{-1}\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right)$ per $a = 0$

$$\begin{cases} 0x = 0 \\ y - t = 1 \\ 2x + 0z = 1 \end{cases}$$

$$\Rightarrow \begin{cases} y = t + 1 \\ x = \frac{1}{2} \end{cases}$$

$$\begin{bmatrix} \frac{1}{2} \\ t+1 \\ z \\ t \end{bmatrix} = f^{-1}\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right)$$

$$f^{-1}\left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \\ 0 \end{pmatrix} + \underbrace{t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{Ker } A} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ 0 \\ 0 \end{pmatrix} + \text{Ker } f$$

$$= (1/2, 1, 0, 0)^T + t(0, 1, 0, 1)^T + z(0, 0, 0, 1)^T$$